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Efficient Well Intervention Eliminated Liner and Tie Back with Efficient Single Casing Expediting Well Delivery

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Abstract

This paper outlines an innovative approach to optimize workover re-entry wells eliminating the liner and tie-back operations by addressing challenges that had placed this strategy as not suitable in the previous history of the onshore operations around the area. The main concern deemed impossible was cementing a 7" Production Casing from the landing point to the surface safely and successfully. This strategy is essential for the development of the field and to efficiently deliver the well count on time. The project started by designing and selecting the casing materials, required auxiliary equipment and mitigating all the risks associated with the equivalent circulating density (ECD) during drilling and cementing and refining operational procedures, the projected provided a more operationally efficient well, in a shorter time, lower cost and a more integral well with not weak points as the tie-back connections and the stage tool ports. This technique eliminates the need for running and cementing both a 7" liner and a tieback, streamlining the process while ensuring uncompromised well isolation.

Hydraulic simulation, drawing from available field data, was employed to assess acceptable equivalent circulating densities (ECDs) and pressures without risking losses. These simulations guided the refinement of cement slurry designs. A specialized lightweight cement system was suggested to address ECD margin limitations and challenges linked with conventional lightweight cement. Additionally, an expandable gas-tight system was introduced to prevent zonal communication and micro annulus in between the different reservoirs. Further enhancements included the introduction of sealing/bridging spacer to ensure effective mud removal along with the optimization of pumping rates to minimize losses during displacement.

Optimized pumping rates during drilling, well bore cleaning and conditioning and cementing operations were guided by simulation data to ensure that dynamic pressures remained safely below fracture pressure thresholds. Additionally, it was decided to include higher density mud to assist in reducing surface pressures during the displacement. The utilization of a specialized lightweight slurry proved effective in decreasing hydrostatic pressure of the fluid column, thereby preventing lost circulation incidents and guaranteeing a positive indication of cement return to the surface.

The drilling fluids conditioning, the spacers train and the exceptional rheology of the slurry facilitated improved mud displacement and enhanced cement coverage during placement. Furthermore, a higher solid volume fraction (SVF) in the cement contributed to reduced porosity and permeability, ensuring long-term zonal isolation and casing protection, thereby minimizing maintenance requirements and extending the well's lifespan, especially in critical reservoirs.

The expandable property of the set cement mitigates for any potential micro annulus development, helping to avoid sustained casing pressure (SCP) complications throughout the well's operational life, thus reducing the need for further remedial cementing operations.

Moreover, a specialized spacer system efficiently aided in mud removal and demonstrated excellent bonding, as confirmed by cement evaluation logs (CEL) conducted across the 7" casing. This comprehensive design approach effectively eliminated additional operational costs and rig time by circumventing the necessity for further remedial interventions.

This paper aims to showcase the utilization of 7" production casing in conjunction with specialized cement design as an efficient substitute for conventional liners and tiebacks. The goal is to reduce rig costs and save time during workover re-entry wells, while also minimizing the need for future costly remedial works.

Introduction

Oilwell construction requires large capital expenditures and investments. In occasions planned budgets are exhausted before project completion, due to difficulties after the original planning, unexpected situations or challenges during the execution phase. Currently the oil industry globally is dedicating extreme efforts to reduce costs, carbon footprint and ultimately increase hydrocarbon recovery, safely and at a lower cost.

This study covers the deployment of an efficiency project that contributed to the overall cost reduction of the well construction, by reducing the rig days, enhanced well integrity by eliminating weak points in the tubulars and extended the well lifespan by securing zonal isolation across multiple hydrocarbons contained reservoirs.

The engineered approach included a thorough evaluation of current well construction practices and the implementation of alternatives that reduced risks and well maintenance operations, improving overall project efficiency, cost, carbon footprint, and health, safety and environmental (HSE) compliance.

Well architecture and efficiency

The drilling function is always trying to get better wells more efficiently, by looking at the typical operation for the major projects, identifying some areas with room for improvement that have detonated very valuable projects as the implementation of a full 7in production casing for gas lift wells, replacing typical liner and tie-back operations.

Advantages of Using 7-inch Casing

Structural Integrity and Simplicity

- A full 7" casing string from bottom to surface provides robust structural support and eliminates the need for liner hangers or tie-back seals, which are potential weak points and more risky, complex operations.
- It simplifies the well architecture, reducing the number of interfaces and mechanical risks while eliminating the unintentional cement left in the lower section/completion of the well during tie-back operations.

Operational Efficiency

- Deploying a 7" casing instead of a liner and tie-back has saved up to 4 rig days in onshore operations.
- This translates to significant cost savings and reduced exposure to operational risks and emissions.

Improved Integrity Assurance

- Full casing allows for comprehensive pressure testing and integrity validation without relying on seal stem tests or multi-stage cementing hardware, which can be sensitive to overpressure and mechanical and operational failures.

Reduced Risk of SAP B (Sustained Annulus Pressure)

- An additional highlight is that SAP B issues often arise from leak paths behind tie-back seals, this is eliminated with the casing to surface.
- Using full casing mitigates this by removing the tie-back interface altogether.

Challenges with 7-inch Liner and Tie-Back

Complex Testing Requirements

- Tie-back seals require separate pressure testing post-cementing, often limited to 1,500 psi to avoid premature activation of cementing hardware.
- Testing protocols vary and must be customized per well, adding complexity and potential delays.

Integrity Risks

- Leak paths have been observed behind tie-back seals, especially when production packers are set in the new tie-back casing instead of the old liner.
- This raises concerns about secondary barrier compliance and long-term well integrity.

Operational Overhead

- The need for pre-job meetings, simulations, centralizer spacing, and contingency planning for tie-back operations to reduce the risk associated with these operations as cement left in pipes and tool failures is considered high.
- These steps add to the planning and execution burden compared to running a single casing string.

Cost and Duration

- While liners are generally cheaper in material, the added duration and complexity of tie-back operations can offset these savings. Global teams have flagged this as a business plan constraint for when tubulars need to extend to surface.

Overview of the two options

Casing offers superior structural support, better well control, and contamination protection and not necessarily at higher cost.

Liners are more flexible and cost-effective but introduce weak points, limited support and when a tieback is required the saving are offset and the total cost is higher than a single-cased section.

Both well construction options require primary cementing operations to create a hydraulic seal between the tubular and the drilled formations. Liners cementing represent a risk in terms of the back pressure created in the liner hanger and previous casing, where annular gaps can be as low as 0.16 inches, creating a pressure drop above the restriction and generating an overpressure below that can induce losses while cementing across weak, depleted and natural fractured formations, risking the cement coverage. Casing cementing doesn't have this kind of restriction, anyhow, generally the top of cement is designed at lower depth or surface, increasing the overall hydrostatic and friction pressure (ECD).

To overcome these challenges a specialized lightweight cement system was implemented that combined with solids of low specific gravity with an engineered particle size distribution allowed higher solid volume fraction (SVF), improving the set cement mechanical properties, expandable and gas-tight properties add assurance to mitigate gas migration during the critical hydration period and preventing micro-annulus that could be created by the cement bulk shrinkage during the cement setting. Additionally, a specialized cement spacer with shielding properties was introduced to mitigate the pressure transmission to the formation allowing higher ECD's during cementation without inducing losses.

Cementing Design

Primary cementing operations across long intervals and narrow annular spaces presents additional challenges to the cementing design and challenges the likelihood of success. ECD's during cementing are typically higher than those observed during drilling. This is due to the use of higher-density fluids, increased SVF's, and restricted annular clearances—all contributing to elevated hydrostatic and frictional pressures. The sum of these pressures is referred to as the ECD.

An increase in ECD during cementing can induce losses in weak formations within the interval, compromising cement coverage and zonal isolation. This may lead to unwanted annular flow to uncemented permeable zones or even to surface, resulting in hydrocarbon losses, contamination of underground aquifers, and increased risk of annular pressure build-up (APB) or sustained casing pressure (SCP). These issues can also accelerate casing corrosion early in the well's life, increasing maintenance costs, requiring workover interventions, and ultimately reducing the well's lifespan and asset profitability.

To mitigate these risks, a robust engineering approach is essential. This includes a detailed cementing design, specialized fluid systems, hydraulic simulations, and the implementation of both engineered and field-proven best practices to ensure a successful operation. The engineered cementing strategy applied for a 7-in horizontal production casing will be outlined in the following section.

Fluids Design

Specialized Spacer Design. Conventional cementing spacers serve several critical functions: they separate the drilling fluid from the cement slurry to prevent the formation of incompatible fluids that could jeopardize the cementing operation, assist in removing the drilling fluid to prepare the annular surfaces for effective cement bonding, and must be compatible with both the drilling fluid and the cement slurry. In cases where non-aqueous drilling fluids are used, surfactants and solvents are often added to the spacer to water-wet the casing and formation surfaces.

Chemical washes can also be pumped ahead of the cementing spacer to thin, dilute, and disperse residual drilling fluid. These fluids are typically low-density, either water- or oil-based, and may help reduce ECD during cement placement. Due to their Newtonian behavior, chemical washes can achieve turbulent flow—often the preferred flow regime—at relatively low pump rates.

For weighted spacers, the alternative to turbulent flow is **effective laminar flow (ELF)**. To achieve ELF, the following criteria must be met:

- Spacer density should be at least 10% higher than the fluid being displaced.
- Friction pressure must be at least 20% higher than that of the fluid being displaced.
- Fluids must flow completely around the casing in the annular space.
- Annular fluid velocities should be uniform throughout the annulus.

In some field applications, especially due to wellbore constraints, the ELF criteria cannot be fully met. Limitations related to maximum allowable ECD, fluid density, and rheological properties may require the spacer system to be designed with characteristics closer to the drilling fluid being displaced.

An additional challenge in the case study was **casing stand-off (CSO)**. Achieving adequate centralization in horizontal casing cementing is critical and depends on using enough centralizers to meet the required CSO. Factors such as dogleg severity, tubular stiffness, and formation properties must be considered during the centralization simulation and placement planning.

To improve mud removal and provide formation strengthening, a **specialized spacer system** was implemented. This system consists of a mixture of long-chain polymers and flexible solid particles of various shapes and sizes. These solids are activated upon contact with the formation under differential pressure (i.e., overbalance conditions), forming a thin, flexible, impermeable shield on the formation face. This shield restricts pressure transmission, seals existing losses, and prevents further fluid invasion or loss, thereby minimizing formation damage.

Once the well transitions to underbalance conditions, the solids in the specialized spacer can be easily removed if required.

Figure 1 illustrates the sequence and mechanism of action of the shielding particles: from pumping, displacement into the annulus, to activation upon contact with the formation, ultimately forming an impermeable barrier.

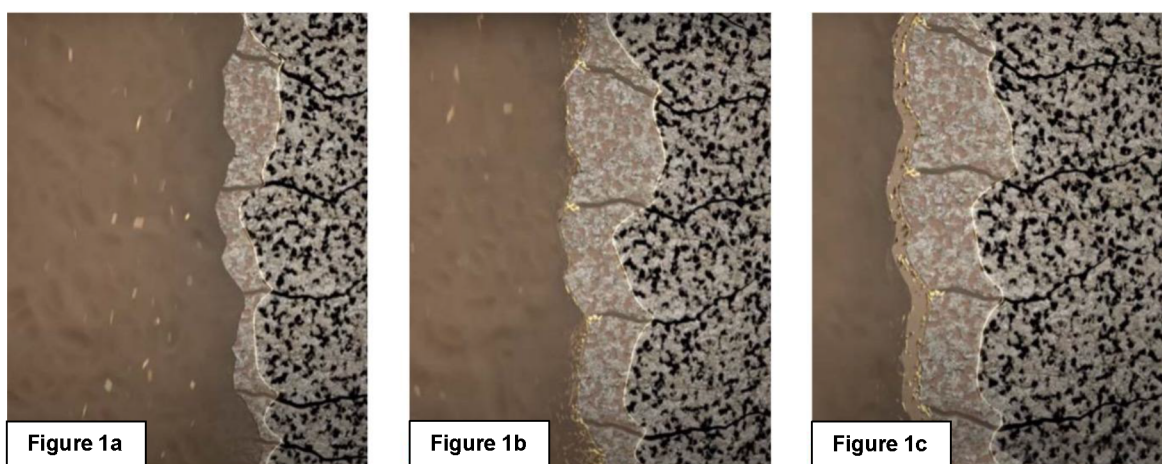


Figure 1—Specialized cementing spacer solids particles disperse in the fluid (1a). Solids start plugging the pores in the formation face (1b). Thin impermeable barrier is created, isolating the formation to be cemented, preventing fluid losses and formation damage (1c).

The specialized spacer system allows for ECD's slightly above the formation pressure, enabling cement slurries to cover longer annular intervals and preventing cement fallback. The solids in the spacer, combined with annular fluid velocity, exert an erosive effect on the annular surfaces, assisting in the removal of difficult-to-displace gelled mud.

While the primary function of the spacer is to shield the formation, thereby mitigating and preventing losses—it also helps reduce cement slurry filtrate loss. This contributes to improved cement setting by

maintaining the correct solid-to-liquid ratio, allowing the cement to set with mechanical properties closer to the laboratory design.

Figure 2 presents a comparison between two cementing scenarios:

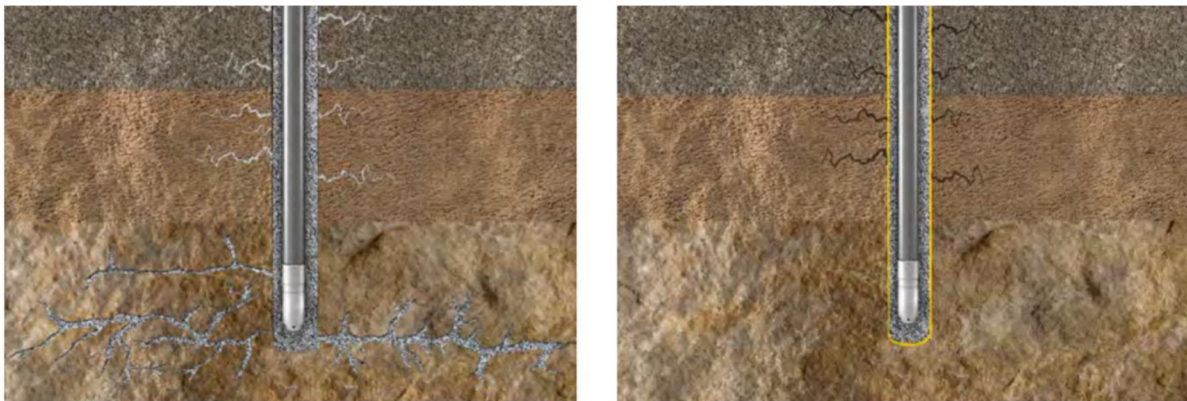


Figure 2—Cementing operations without specialized spacer system shielding the formation (left). Cementing operation with the formation shield, fluid loss and fibers (right)

- On the **left**, a cement job without formation shielding shows fluid invasion into natural fractured and weak or depleted formations.
- On the **right**, the presence of shielding spacers, fluid loss additives and fibers added to the slurry systems prevents fluid migration and formation damage while enabling extended annular cement coverage.

The solids particles contained in the specialized spacer system has a plugging capacity up to 400 microns. This can be extended up to 3,000 microns by pairing it with a natural occurring environmentally friendly loss circulation material (LCM). To maximize the fluid plugging efficiency, the spacer must be designed with an adequate Yp (surface and downhole) and a laboratory slot test to be performed to confirm fluid stability and plugging performance.

To validate the spacer system laboratory tests were performed as per API RP-10B, additionally the following tests were also performed:

Spacer fluidity and mixability. To assess the rheology and mixability of the specialized spacer, a laboratory sample is prepared by combining the spacer with the LCM. The mixture is poured into a sample cup and blended using a tabletop paddle mixer, which replicates field mixing equipment.

In Figure 3 (left), a 12 lb/gal spacer sample paired with 25 lb/bbl. of LCM with 3,000-micron plugging capability is shown during pourability evaluation by transferring the mixed sample between cups and inspecting for signs of solids agglomeration or reduced flow. Such issues could pose operational risks in the field. In Figure 3 (right) continuous flow is observed on the tested sample while pouring the fluid into a sample cup.

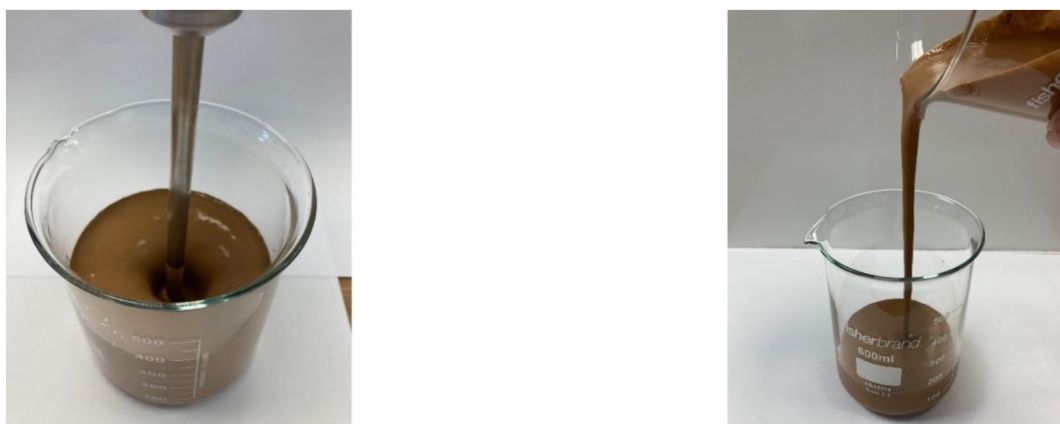


Figure 3—Evaluation of the mixability and rheological behavior of the specialized spacer paired with the LCM

Spacer static and dynamic stability test. During cementing operations, fluids are subjected to both static and dynamic conditions; therefore, stability testing is conducted to ensure performance under these variations. The stability of the specialized spacer is evaluated by placing a laboratory sample into an HPHT consistometer and conditioning it to downhole temperature and pressure. The resulting consistency chart is analyzed for any increase in consistency, which may indicate the development of high gels, or for any decrease, which could suggest solids settling within the spacer.

Figure 4 presents the consistency profiles of 12 lb/gal and 14 lb/gal specialized spacer systems; each paired with 25 lb/bbl. of environmentally friendly with 3,000 microns plugging capacity LCM. Both systems demonstrate stable behavior under simulated downhole conditions.

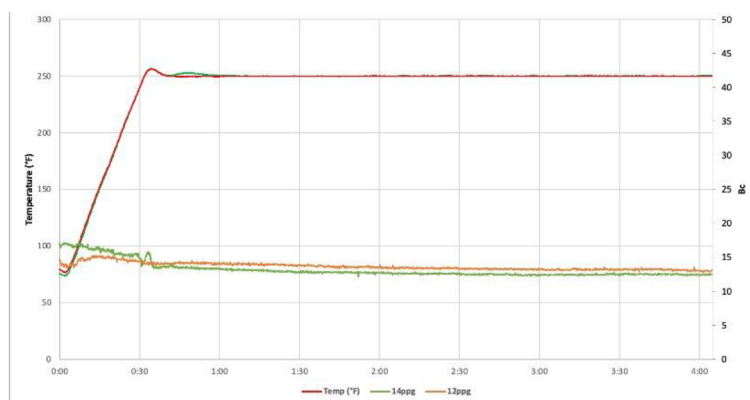


Figure 4—HPHT consistometer chart showing a 12 lb/gal (orange) and 14 lb/gal specialized spacer system mixed with 25 lb/gal LCM

Upon completion of the dynamic stability test, the slurry cup and paddles from the HPHT consistometer are visually inspected for any signs of solids agglomeration or settling at the bottom of the cell. As shown in Figure 5, neither component exhibits evidence of such issues. To further verify stability, the spacer density is measured at both the top and bottom of the sample, confirming the absence of significant variation.



Figure 5—Visual inspection of the paddles and the bottom cap of the slurry cup after the completion of the spacer dynamic stability test

Upon successful completion of the dynamic stability test, the fluid sample is poured into a 250 ml graduated cylinder and left undisturbed for two hours to conduct the static stability test. A visual inspection is then performed to confirm fluid stability, ensuring that no free water is present on the surface of the sample. An example of this test setup is shown in [Figure 6](#).

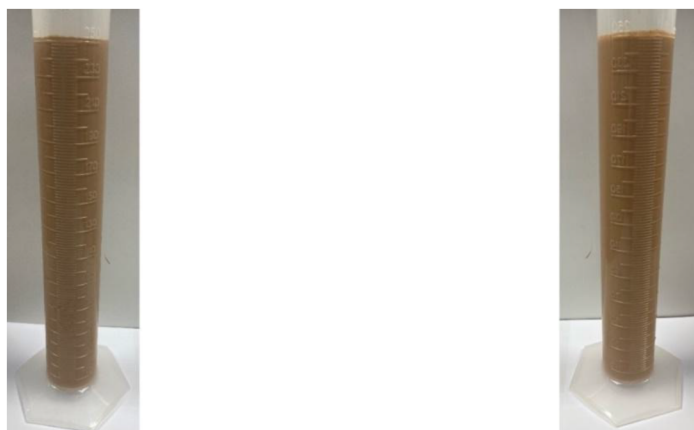


Figure 6—Static stability test performed for a 12 lb/gal spacer system. Sample at the start of the test (left) and after 2 hrs (right)

Similar to the dynamic test, at the end of the static stability test, fluid samples are collected from the top, middle, and bottom of the graduated cylinder using a syringe. Each sample is then compared with the design fluid density. [Table 1](#) presents the results for two spacer samples—12 lb/gal and 14 lb/gal—each paired with 25 lb/bbl. of environmentally friendly LCM (3,000 microns plugging capability). In both cases, the density variation was less than 1.5% and within the acceptable range.

Table 1—Static Stability test results for a 12 lb/gal and 14 lb/gal specialized spacer system paired with 25 lb/bbl. environmentally friendly LCM

Segment	Density (ppg)	Density (ppg)
Baseline	12	14
Top	11.9	13.8
Middle	12	14
Bottom	12	14.1

These results confirm that the specialized spacer system can maintain large solids load in suspension under both surface and downhole conditions, preventing fluid instability that could lead to solids agglomeration or sagging. Such instability could potentially plug restrictions within the system.

Spacer plugging capacity test – slot test. To validate the plugging capacity and efficiency of the specialized spacer design, slot tests were performed using a particle plugging apparatus (PPA). The PPA can be operated either hydraulically or with gas. For each test, the sample is poured into the testing cell, which contains the selected slot installed at the bottom of the cell. A pressure of 1,000 psi is applied from the top of the chamber while the bottom valve remains closed.

Once pressure is established, the bottom valve is gradually opened, and the initial fluid spurt is collected in a pre-weighed plastic container. The spurt volume is weighed, and the test continues for 30 minutes, after which the total fluid loss (TFL) is measured. Figure 7 shows the main components of the gas-operated PPA and the reference slots ranging from 1,000 to 3,000 microns.

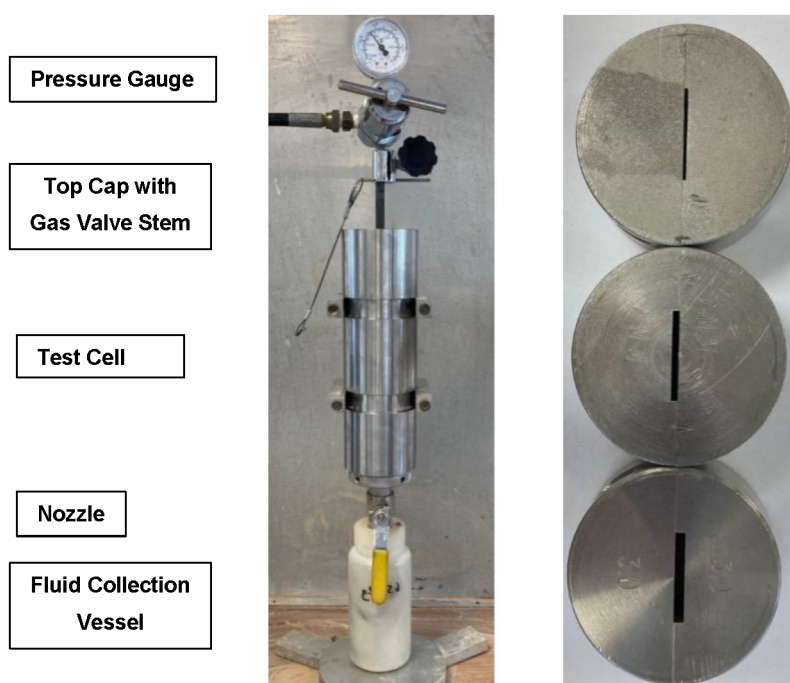


Figure 7—Gas-operated Particle Plugging Apparatus (PPA) showing main components and reference slots ranging from 1,000 to 3,000 microns.

Upon completion of the test, the applied pressure is released, and the TFL is measured both in grams and as a percentage of the initial fluid sample. The filter cake formed on top of the slot is then visually inspected. Figure 8 shows the filter cake formed on a 3,000-micron slot after a 1,000-psi differential pressure slot test using a 12 lb/gal enhanced spacer paired with 40 lb/bbl. LCM.

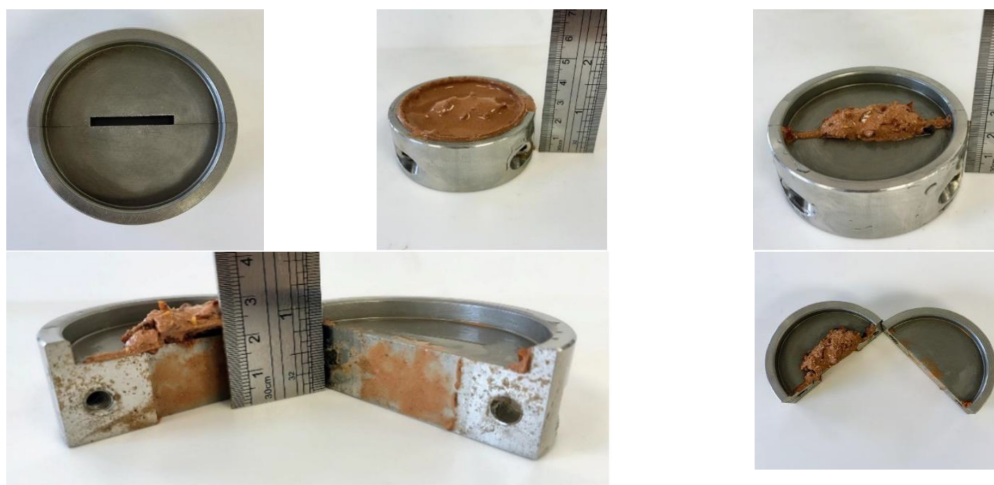


Figure 8—3,000 microns slot and the formed filter cake in a 12 lb/gal enhanced spacer paired with 40 lb/bbl. of LCM

Slurries Design

The engineered completion proposal required full annular coverage with cement, with the top of cement (TOC) at surface. This option substantially increased the equivalent circulating density (ECD) during pumping and the final hydrostatic pressure, creating a high risk of cement fallback. The conventional approach is to pump a low-density "lead slurry" followed by a tail slurry at API Class G density (15.8 lb/gal) or higher. Lead slurry generally has a low solids volume fraction (SVF), which translates into higher porosity and permeability and lower compressive strength.

Conventional extender particles are typically added to the lead slurry design to increase SVF. However, these particles tend to crush under downhole conditions, which increases the slurry density and reduces the SVF. Another challenge is that the OH contained gas-bearing reservoirs, requiring that the slurry contained both low API fluid loss and rapid transition during setting to mitigate gas migration in the well.

To address these challenges, a non-crushable lead slurry was introduced. This design prevents density increase under downhole conditions while maintaining the planned SVF. It also provides good API fluid loss in between 30 -50 cc and rapid gel development during setting. Compressive strength was significantly improved, reaching ~1,100 psi by destructive testing and an average of 2,000 psi at 24 hours in the ultrasonic compressive strength analyzer.

Overall, the non-crushable slurry design reduced hydrostatic pressure, preventing lost circulation or formation breakdown. It lowered slurry water content (higher SVF), which enhanced compressive strength, reduced permeability, and limited fluid invasion into the cement matrix. Faster gel development and early compressive strength shortened the waiting-on-cement (WOC) time. These improvements in lead slurry properties helped prevent unwanted fluid migration between zones or to surface (SCP), improved cement bonding and zonal isolation, and reduced the need for costly remedial operations, ultimately saving rig time.

Other considerations

Centralization. To achieve a minimum 70% stand-off, two rigid centralizers were installed per casing joint in the highly deviated open-hole section, and one centralizer every two joints in the casing-to-casing annulus. This configuration achieved an average stand-off of 82%. In the last two joints near the surface, two centralizers were placed per joint.

Wellbore cleanout & Pre-flushes. In addition to the shielding spacer and to optimize mud removal, a minimum of 1.5 circulation cycles were completed, or until the returns were free of cuttings. Circulation was initiated at a low rate to break the gelled mud, then gradually increased to reach the maximum anticipated pressure during cementing operations (if no losses were expected). A high-viscosity pill was included to

aid in transporting larger particles to the surface. A 200-bbl volume of low-viscosity mud was pumped as a pre-flush ahead of the shielding spacer.

Water- and oil-based chemical washes were also considered. These Newtonian fluids can readily achieve turbulent flow under standard pumping rates, improving mud dilution and overall removal. Special attention was given to always maintaining overbalance pressure to prevent formation fluid influx into the wellbore.

Operational best practices.

- Both the non-crushable lead slurry system and the tail slurry were batch-mixed to avoid density variations that could affect pumpability and set cement properties.
- Five casing joints were kept as a shoe track (the space between the casing shoe and the casing collar) to minimize the risk of cement contamination at the landing point.
- The casing was reciprocated with 10-ft strokes during mud circulation to improve wellbore conditioning.

Case Studies

Well A

The first case study involves running 7-in casing to 9,548 ft MD / 8,753 ft TVD in an 8 ½-in open hole. The section was drilled with 10.8 lb/gal OBM with minor dynamic losses at TD, the casing was successfully run to the planned shoe depth. Given the length of the casing and calculation for contamination form, the bottom wiper plug a 4 joint shoe track was designed covering 160 ft and a volume capacity of 5.9 bbl. After overcoming the challenge to run casing to TD successfully and without inducing losses, the casing had two centralizers per joint on the open hole and reducing to one centralizer per joint in the vertical section of the casing, the next step was to circulate to condition the drilling fluid, clean the open hole and prepare the wellbore for cementing operations again without inducing losses, so several very detailed simulations and a field pumping rate calibration allowed us to fully circulate around two cycles with the recommended parameters while rotating as per simulations including a high viscosity pill and a tow slugs of low viscosity drilling fluid. The main challenges were managing high ECDs during cementing and bringing TOC to surface, with maximum pressure approaching the prognosed fracture pressure (Figure 9), during the cementing operations again the wellbore clean out and fluids dynamics were designed to ensure effective mud removal and casing stand-off in the highly deviated section. With the casing at bottom, circulation was initiated at 2–3 bpm and gradually increased to a maximum of 6 bpm until completing 1.5 cycles plus the HIVIS and LVIS fluids, after which the mud was conditioned and confirmed clean of cuttings at the surface solids-control equipment.

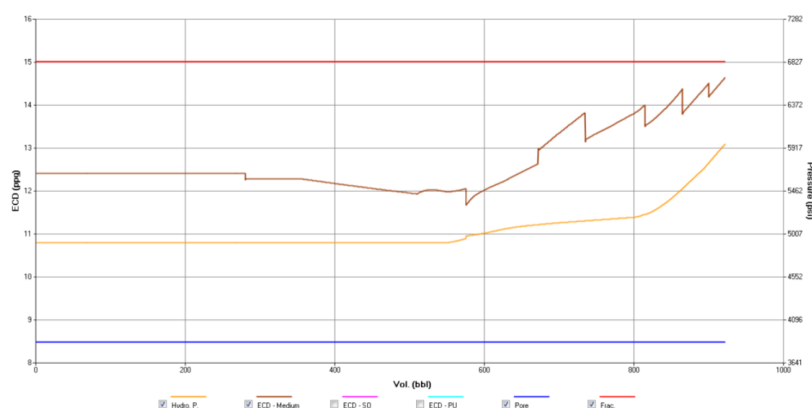


Figure 9—Well A: 7-in casing equivalent circulating density at 9,548 ft.

Once the wellbore was clean, the design was simulated for the rig pumped 200 bbl. of low-viscosity mud, switched to the cement unit, and began the cementing sequence:

- 80 bbl. of 12 lb/gal specialized engineered spacer
- Bottom plug release
- 185 bbl. of batch-mixed 11 lb/gal non-crushable lead slurry
- 130 bbl. of batch-mixed 15.8 lb/gal expandable gas-tight tail slurry
- Top plug release
- 10 bbl. of conventional weighted spacer to start the displacement

After confirming the top plug release, displacement continued until bumping the plug. Pressure was raised to 3,000 psi and held for 10 minutes to test the casing, then bled back to the cement unit displacement tanks to verify float equipment integrity. Pumping rates, densities, and pressures in the design and simulation for the cementing job are shown in Figure 10.

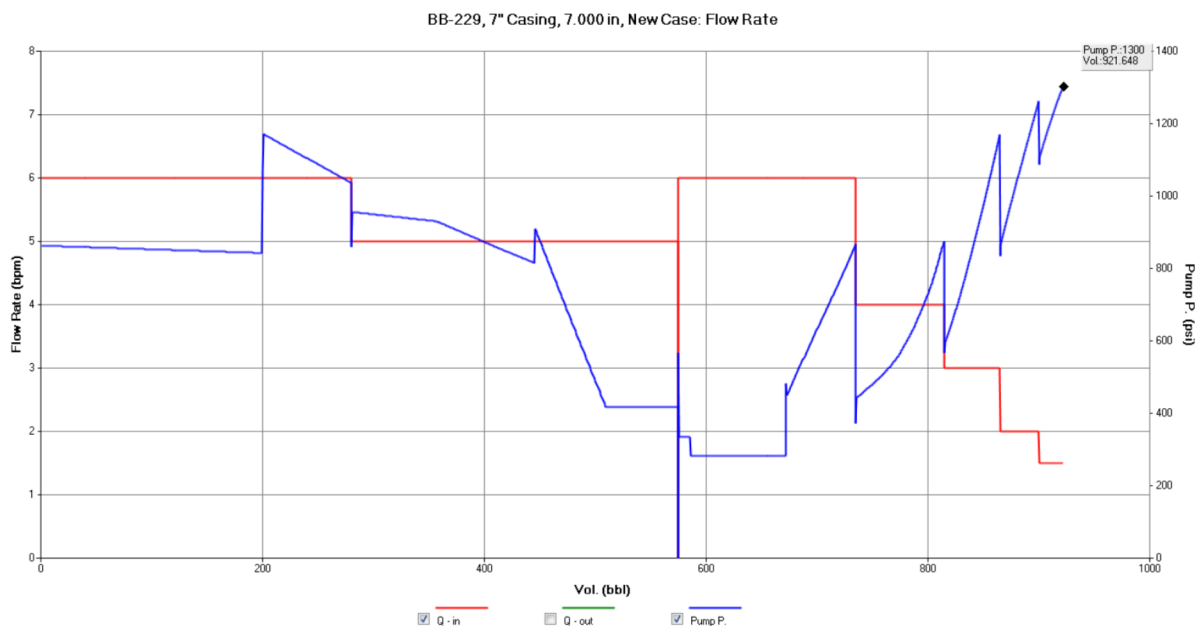


Figure 10—Well A: Job design and parameters simulated for the 7-in casing cementing operation

No losses were reported during the cementing operation, and good cement returns were observed at surface with the level remaining static. The cement evaluation log confirmed good bonding across the target interval, successfully meeting the cementing objectives.

The main design parameters and simulation were analyzed, and several mitigations were implemented to mitigate for the apparent risks that in the past were the reason for never attempting to deploy an efficient 7in casing from LP to surface, below in Table 2 the analytical results are presented and the field job monitoring and recording chart Figure 11.

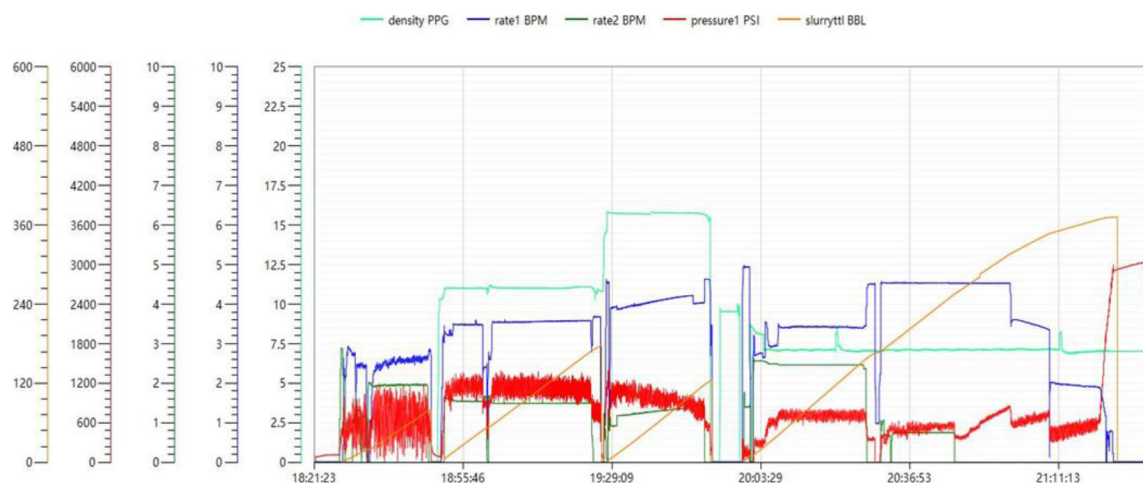


Figure 11—Well A field job monitoring and recording chart in the field.

Table 2—Well A pre-job preparations, simulations, analysis and risk assessment of key job parameters.

	Description	Value	Notes
1	Total Depth (MD /TVD)	9,548' /8753'	Depths as per the last directional plan. Rev_B0 Report any changes or if losses found during drilling to DE.
2	Top of Casing	Surface	
3	The highest Pore pressure exposed is Zone A	4230 PSI	
4	Minimum static overbalance achieved <u>on Zone A</u>	571 PSI	
5	Time to reach Minimum static overbalance	Begin of Job	
6	Pumping volume to reach Minimum static overbalance	Begin of Job	
7	Fracture Pressure Gradient	0.78 psi/ft.	
8	Dynamic margin to fracture at liner shoe	162 PSI	
9	Confirmation that safe margins will remain as above by not having made changes to the well design (e.g. casing / liner tally) and by planning pumping rates and volumes as per original plan.		Report any changes or differences to DE.
10	Confirmation that safe margins will remain as above by having reviewed and checked independently that the rig team has the same conditions for well design. <u>In particular pore</u> pressures are as per program, trajectory changes or new zones with different pressures are updated and so are MDT results. The planned pumping rate and volume as per original plan.		Deviations from the original program to be reported to DE.

With all the preliminary job design and checks, the job preparations, mobilization and field products laboratory testing started, every single aspect was covered, risk analyzed and tested as per the API guidelines specially the fluids scheduled for the job. Figure 12 A, B and C, presents some of the rigorous testing done during the preparation specially for the newly formulated non-crushable 11.0 ppg cement slurry.

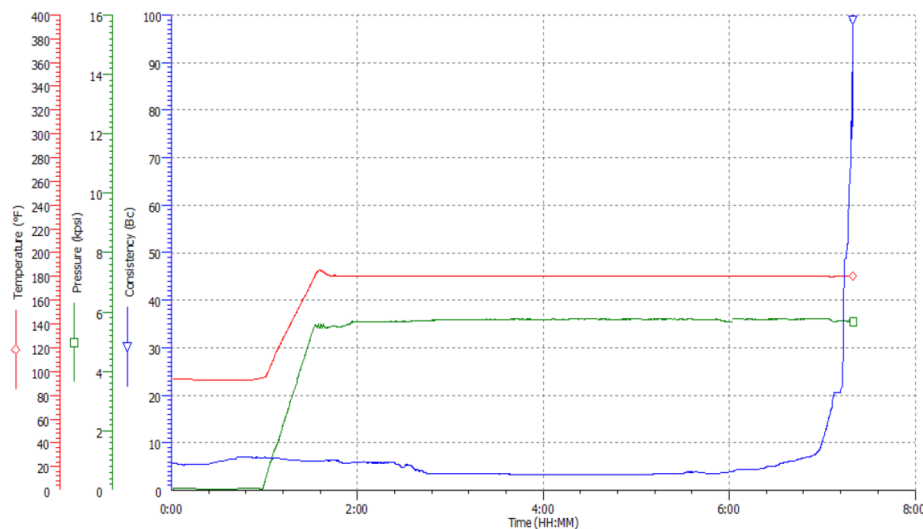


Figure 12 A—7in csg 11.00 ppg thickening time to 100 BC (Full API report was generated for the slurry)

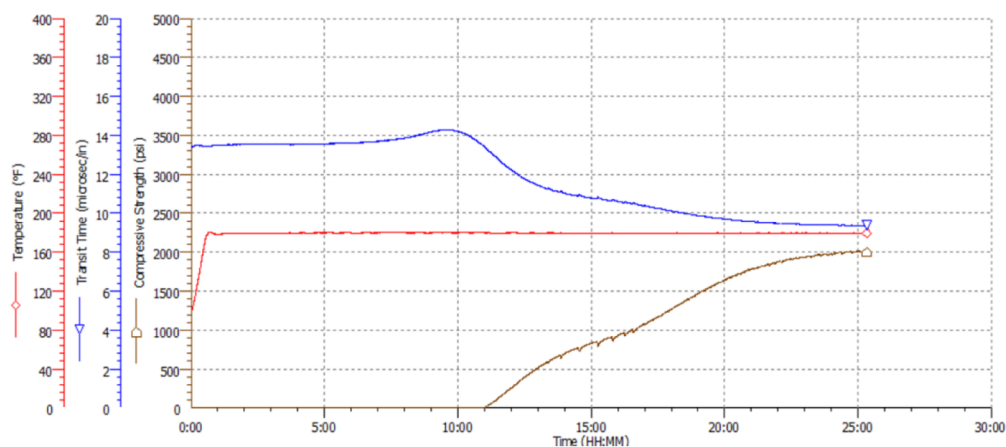


Figure 12 B—7in csg 11.00 ppg compressive strenght type A (less then 14 ppg slurries)

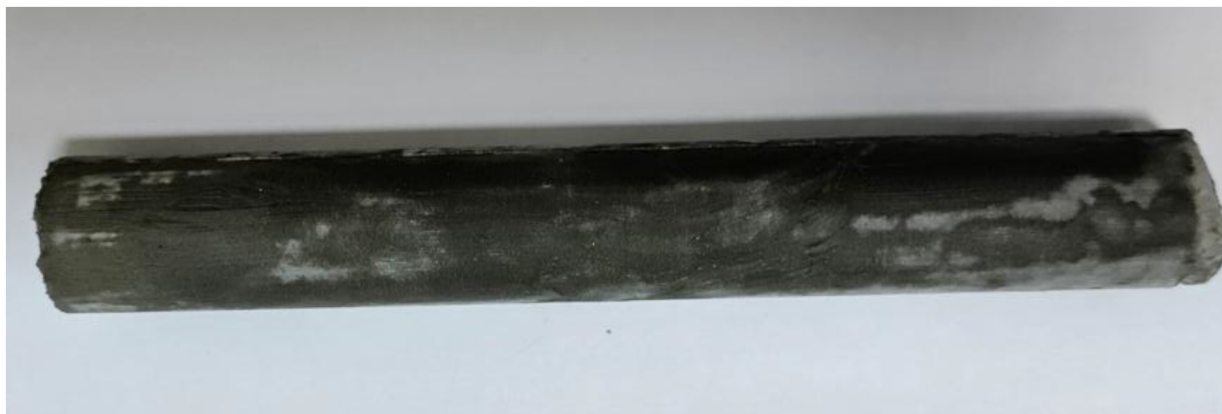


Figure 12 C—7 in csg 11.00 ppg sedimentation test

The field observations were very positive and the well evaluation was a success, all the cementing quality parameters were not only meet but exceed by having additional than recommended pre-job well clean and circulation, fluids returns throughout the cement job, confirmed pure spacer and cement returns on surface, bumped the plug after displacement and successfully pressure testing the casing as required by the code of practice to comply with the integrity requirements [Figure 13A](#) and [B](#).

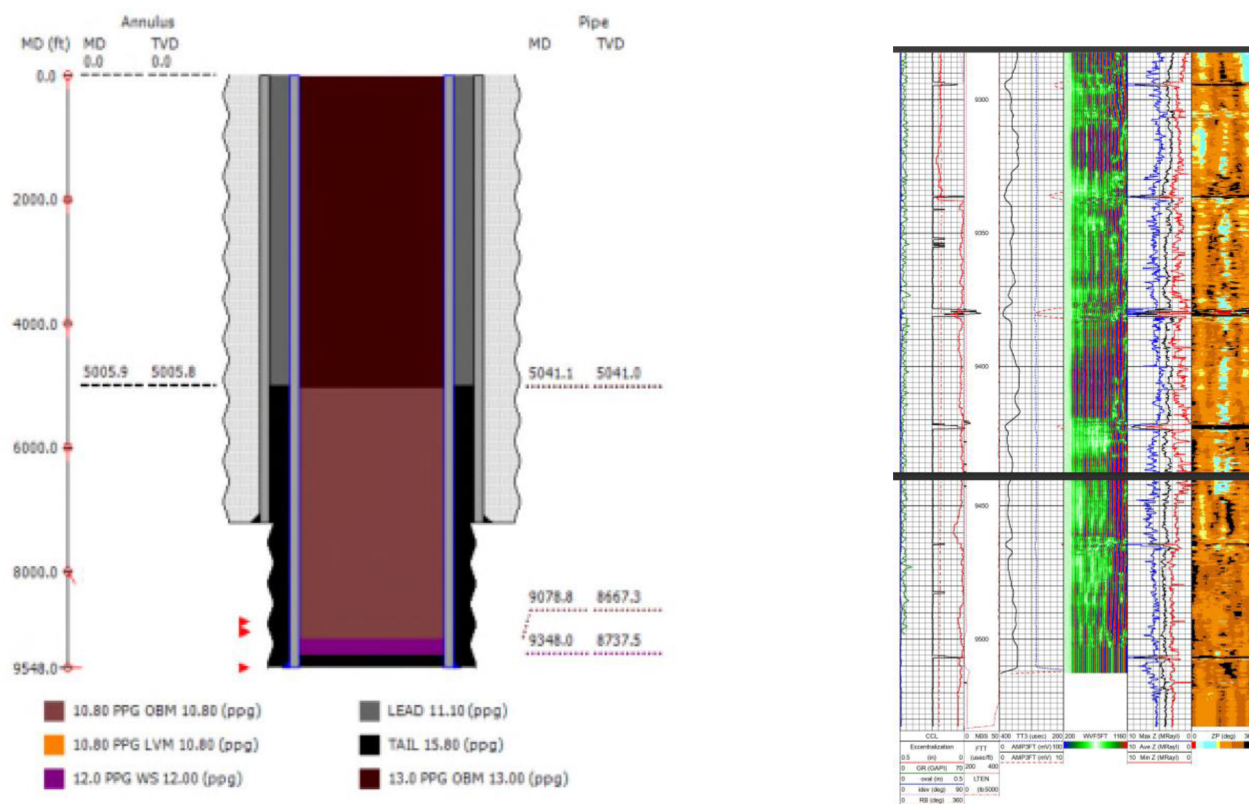


Figure 13A—7in csg final job status with tail cement up to previous csg @ 5005 ft and lead slurry up to surface. Figure 13A. 7in csg final cement evaluation log from LP to surface presenting good isolation across reservoirs, previous casings, and aquifers.

Well B

A 7-in casing was run to 9,518 ft MD / 8,691 ft TVD. Similar to Well A, the 8 ½-in open-hole section was drilled with 10.8 lb/gal OBM without losses, and the casing was successfully run to planned depth. The main challenges for this case study were the proximity of the ECD during pumping to the formation fracture gradient, effective mud removal, the risk of gas migration from gas-bearing formations, and casing centralization in a highly deviated open hole with a maximum deviation of 89.75° Figure 14 shows the well deviation profile, including the horizontal reservoir section, while Figure 15 illustrates the ECD at casing landing depth.

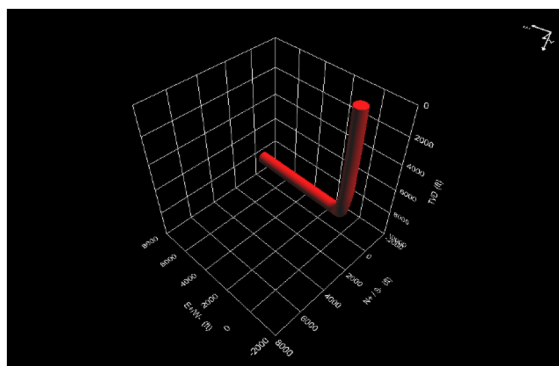


Figure 14—Well B deviation diagram

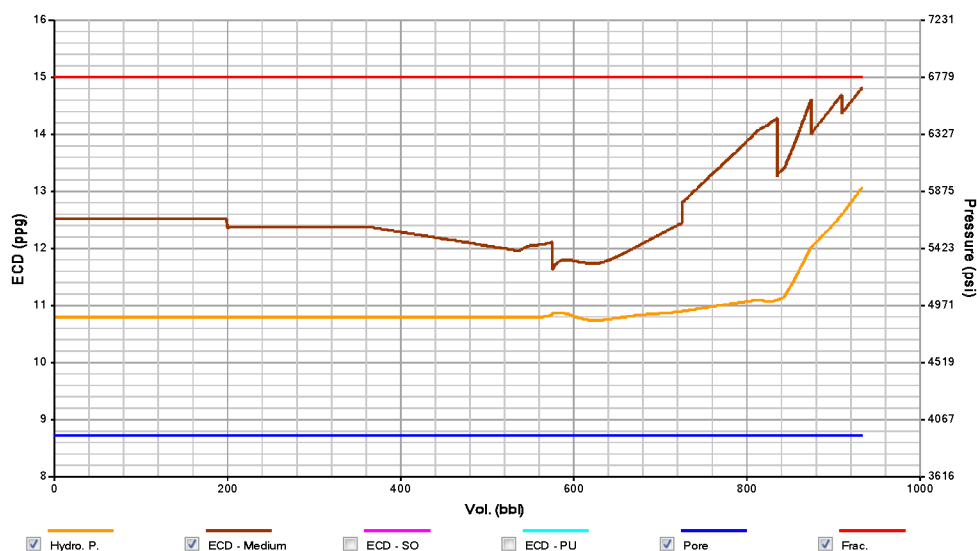


Figure 15—Well B: 7-in casing equivalent circulating density at 9,518 ft

To address these challenges, the pre-flush train included both non-weighted and weighted fluids, comprising water and oil-based chemical washes and an engineered shielding spacer to mitigate pressure transmission to the formation.

To ensure homogeneous fluid flow and cement coverage across the annulus, two rigid centralizers were installed per joint in the highly deviated open hole, and one centralizer every two joints in the cased vertical section.

A non-crushable lead slurry was designed at 11 lb/gal, maintaining stable properties at both surface and downhole conditions. The slurry exhibited zero free fluid, API fluid loss of 50 cc, and stable rheological performance. Its thickening time (TT) and ultrasonic compressive strength (UCA) results are shown in Figures 16 and 17.

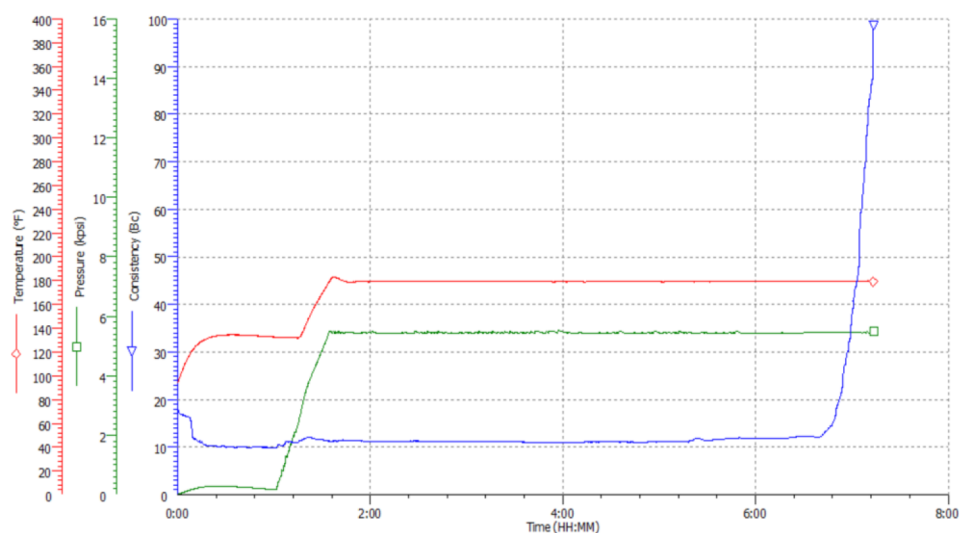


Figure 16—Well B: TT test for the 11 lb/gal non-crushable lead slurry

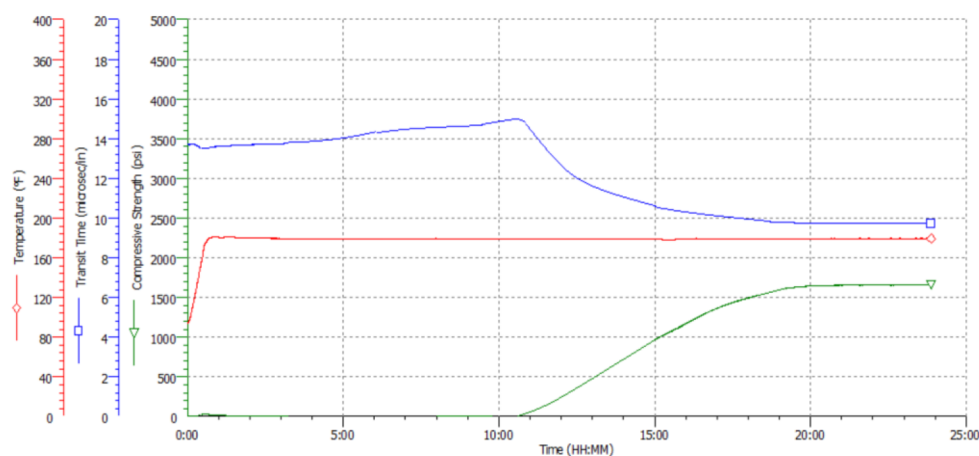


Figure 17—Well B: UCA test for the non-crushable lead slurry

The tail slurry was designed with expandable and gas-tight properties at 15.8 lb/gal, with API fluid loss of 30 cc, zero free fluid, and a right-angle set setting curve. The TT and UCA results are shown in Figures 18 and 19.

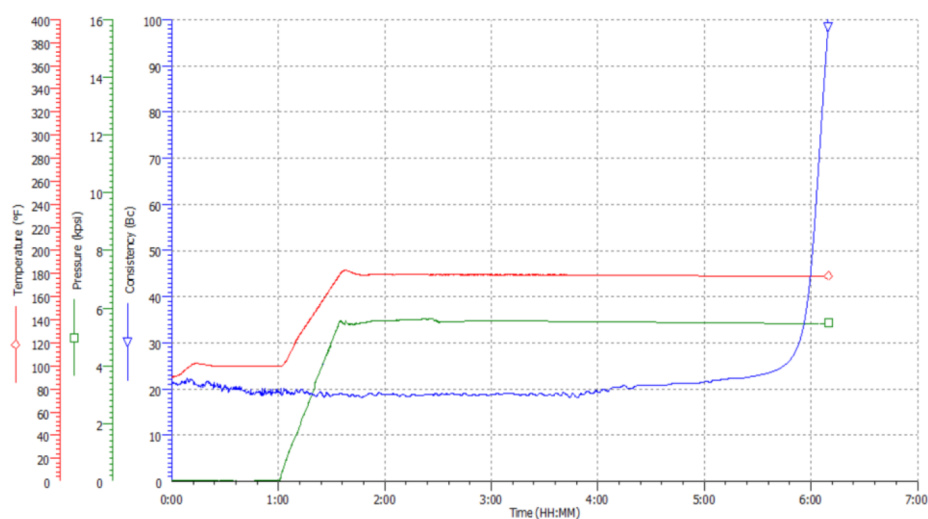


Figure 18—Well B: TT test for the 15.8 lb/gal tail slurry

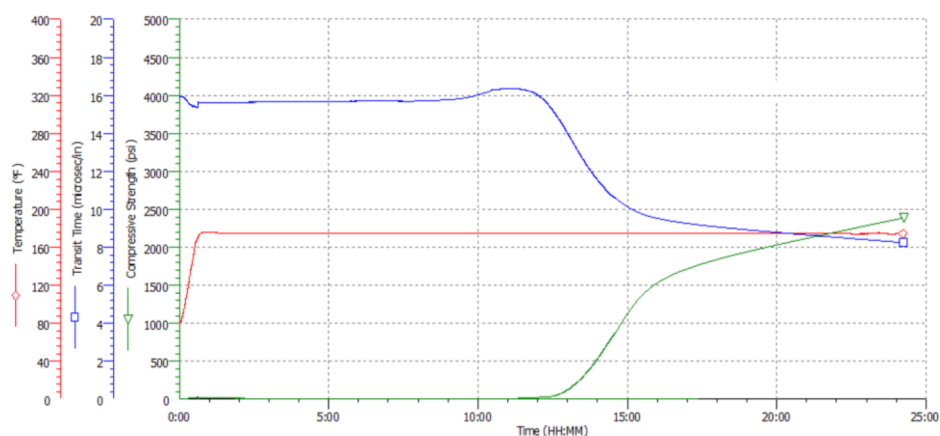


Figure 19—Well B: ultrasonic compressive strength test for the expandable gas-tight tail slurry

With casing at bottom, circulation was initiated for at least 1.5 cycles to ensure the wellbore was free of cuttings. The cementing sequence was as follows:

- 200 bbl. of 10.8 lb/gal low-viscosity mud
- 30 bbl. of oil-based unweighted chemical wash
- 30 bbl. of water-based unweighted chemical wash
- 30 bbl. of engineered shielding spacer
- Bottom plug release
- 165 bbl. of 11 lb/gal non-crushable lead slurry
- 120 bbl. of 15.8 lb/gal expandable gas-tight tail slurry
- Top plug release
- 10 bbl. of 12 lb/gal spacer
- 143 bbl. of 10.8 lb/gal OBM
- 237 bbl. of 13 lb/gal OBM

The plug was bumped, and casing was pressure-tested by increasing pressure 500 psi above displacement pressure. Good cement and spacer returns were observed at surface, and fluid level remained static.

After WOC, a cleaning BHA was run, tagging cement at the landing collar and confirming good cement across the shoe track. An additional 10 ft of new formation was drilled, and the wellbore was displaced with non-damaging fluid. A negative test was successfully performed, after which drilling resumed in the horizontal reservoir section. [Figure 20](#) shows the actual field pressure recorded and monitored. All well in this region are diligently checked for zonal isolation with logging equipment [Figure 21](#).

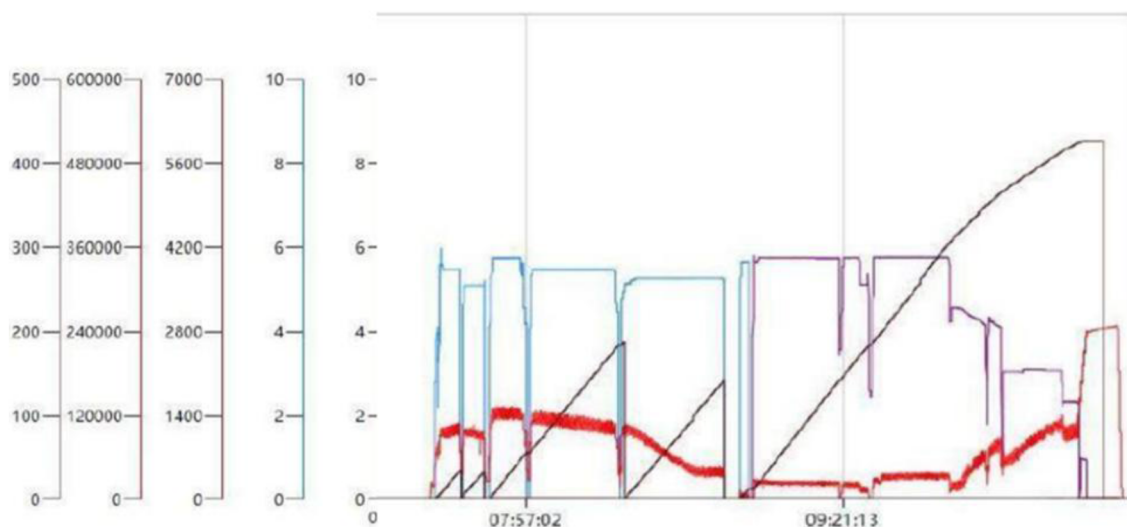


Figure 20—Well B: 7-in casing surface pressure recorded and monitored.

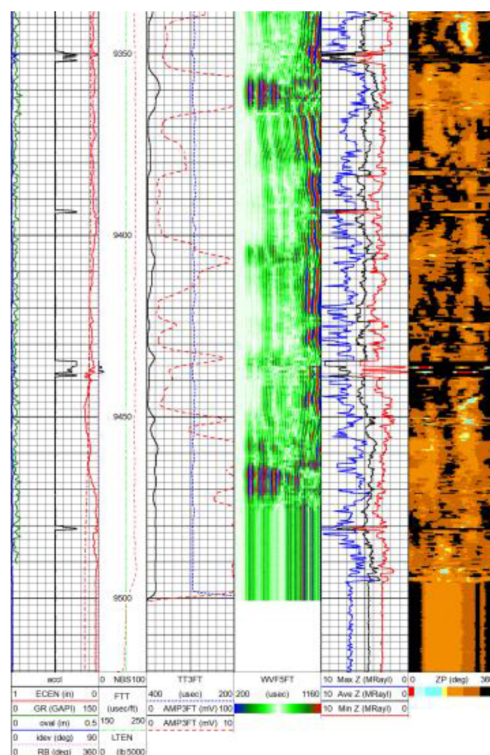


Figure 21—Well B: 7-in casing cement evaluation log extract, showing a similar result as the cement quality parameters.

Conclusions

As is very well known in the oil and gas industry the design of a well includes intensive preparations for different aspects of the well, equipment, drilling, drilling fluids, cementing, completions. The project had a very detailed engineering analysis to validate the concept before the implementation using the latest software's available for accurate simulations along with operational and field observations from previous experiences.

Particularly for the design of this well casing points as per target and the cement job planning, the simulations and analysis were conducted by balancing the hydrostatic pressures of the cement slurries with the pore pressures and formation fracture pressures, as shown in [Figure 22](#).

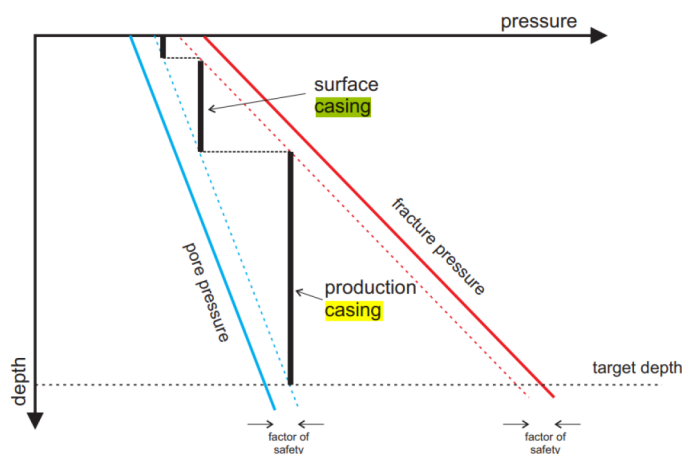


Figure 22—Well B: 7-in casing design and cementing planning and design.

Well design and drilling as well as cement placement in real well conditions is subject to multiple parameters that affect the well construction. While standards and guidelines provide a foundational framework for design, advanced computation fluid dynamics (CFD) simulation tools empower engineers to optimize placement and mitigate for associated risks. These simulations should be designed and customized to exactly the well surface and bottom hole conditions, ensuring well designs are fit for purpose in the most efficient and economical fashion.

Even with the modern tools and with very consistent results, using the latest standards across the oilfield, the well site and final downhole conditions challenges will always require exclusive solutions to that environment to successfully and safely achieve the well requirements eliminating or significantly reducing the generalization of designs and procedures for wells even within the same field or pad.

Adherence to best practices is critical for maintaining well, drilling fluids and cement zonal isolation integrity, including:

- Well architecture design promoting the fluids communication prevention across the wellbore and in between the different zones.
- Ensuring full and radial cement coverage across the wellbore.
- Successful casing tests as per work expected life pressures.
- Overviewing all risks presented by formation, permeability, pressure, fracturing gradients, surface environment, temperate, trajectory, wellbore clean out, fluids specifications to design redundant zonal isolation envelops for each well section.

Specifically this type of wells, deploying a 7-inch casing is recommended over a 7-inch liner with tie-back due to superior well integrity, reduced operational complexity, and proven time and cost saving. This approach is not possible if the primary objective which is mainly to deliver a healthy well is on risk, so it has been demonstrated that engineering analysis for achieving a good cement zonal isolation successfully requires a complete and inclusive approach from all the drilling functions, integrating well engineered cement slurries, optimized mechanical placement through casing centralization and effective mud removal, advanced flow management tools, robust design to withstand operational stresses (pressure, temperature, chemical attacks), and thorough post-job evaluation using tools deployed with e-line, this write up emphasize on proper planning, engineered analysis, proper material selection, best drilling practices and well clean out, intensive job simulations with different software for comparison, personnel competency to allow for appropriate execution and real time decision taking are all critical to ensure zonal isolation ensuring long term well integrity, productivity and longevity preventing any unplanned fluid migration.

Despite the operational challenges, the cementing operations across the 8 ½-in OH and the 7in cased section can be performed successfully with the implementation of the engineered design combined with the operational best practices.

Table 3—Comparative Overview between a cased and a 7-inch liner + Tie-Back well

Criteria	7-inch Casing	7-inch Liner + Tie-Back
Well Integrity	Full string integrity, fewer leak paths	Tie-back seals introduce potential leak paths
Testing Simplicity	Single pressure test for entire string	Requires separate seal stem testing, often limited to 1500 psi
Operational Efficiency	Faster deployment, fewer interfaces	Additional planning, hardware, and execution steps
Rig Time	Saves up to 4 rig days in BAB wells	Longer duration due to tie-back operations
SAP B Risk	Lower risk due to continuous casing	Higher risk from leaking paths behind seals
Cost Consideration	Higher upfront material cost	Lower material cost but offset by longer operations
Barrier Compliance	Easier to validate as secondary barrier	Complex validation due to multiple interfaces

Technical Advantages

The technical advantages are very important because when zonal isolation fails, production or injection efficiency is severely degraded, in some cases the well is lost entirely, or the string is idle until a solution or intervention is scheduled and executed.

No less significant, this type of failures present environmental and safety implications since hydrocarbons or previously injected fluids may flow to the surface unplanned or unnoticed to the nearby aquifers. For this reason it is not sufficient to obtain good zonal isolation, the consequential solutions should provide complete coverage and cover circumferentially the full different casing sections for many years and beyond the expected life of the well.

- Integrity Assurance: Full casing strings eliminate the need for tie-back seals, which have been identified as common leak points in SAP B investigations
- Testing Protocols: Casing allows for comprehensive pressure testing without risking premature activation of cementing hardware
- Operational Streamlining: Casing deployment avoids the need for pre-job simulations, centralizer spacing plans, and contingency hardware
- Field Performance: Internal reports confirm that switching to casing has saved multiple rig days in BAB operations, improving overall efficiency

Recommendations

Onshore operations should prioritize the use of 7-inch casing in applicable wells to:

- Enhance well integrity and reduce leak risks.
- Simplify testing and validation procedures.
- Achieve measurable savings in rig time and operational overhead.

Exceptions may apply in scenarios with specific geological or logistical constraints, which should be evaluated on a case-by-case basis.

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